Every site tells you *which* used car to buy.

None tell you what it will *cost* you.

UsedCarDossier · MGT 409, AI Foundations for Managers

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Yale School of Management · October 2026

`github.com/chanpaca/usedcardossier`

THE PROBLEM

The sticker price is the least informative number in the deal

$11,500	$34,100	3×
WHAT YOU PAY	WHAT IT COSTS YOU	THE GAP

2018 Honda Accord LX, 78,000 miles. The purchase price is under a third of the real cost — and no mainstream site shows the other two thirds.

36 million used cars change hands in the US each year. Second-largest purchase most households make.

THE GAP NOBODY FILLS

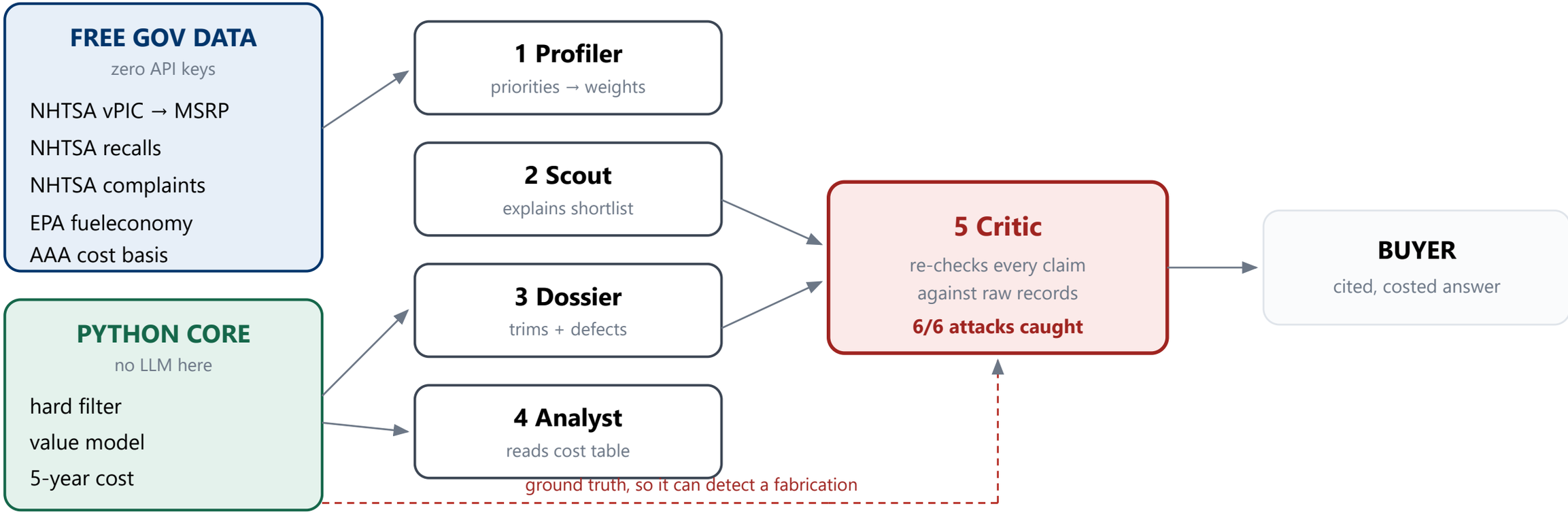
$2,750 hides *inside one model year*

2018 ACCORD TRIM	MPG	FUEL / YR	5-YEAR FUEL
1.5L turbo, CVT	**33**	**$1,950**	**$9,750**
1.5L turbo, 7-spd auto	31	$2,100	$10,500
2.0L turbo, 10-spd auto	27	$2,400	$12,000
2.0L turbo, 6-spd manual	26	$2,500	$12,500

> Pick the wrong trim on styling grounds and you pay **$2,750** extra — a quarter of the purchase price, decided by a line on a window sticker that no shopping funnel asks about.

THE DESIGN DECISION

Python decides and calculates. Agents explain and check.



A model never asked to do arithmetic cannot get the arithmetic wrong.

DEMO

Where $34,100 actually goes

2018 ACCORD, $11,500, 12K MI/YR	PER YEAR	5 YEARS
Depreciation modelled	$940	$4,700
Fuel EPA, exact	$1,560	$7,800
Maintenance, repair, tires	$2,575	$12,877
Insurance	$932	$4,658
Registration and taxes	$813	$4,065
Total cost of ownership	**$6,820**	**$34,100**

Maintenance alone ($12,877) EXCEEDS the purchase price ($11,500). Every line shows the exact basis it was computed from.

THE METHOD THAT MAKES IT WORK

Complaint *counts* measure popularity. Complaint *shares* measure quality.

2018 MIDSIZE SEDAN	COMPLAINTS	ENGINE SHARE	VS SEGMENT
Honda Accord	**1,890**	**13.2%**	**0.88×**
Toyota Camry	**748**	**4.7%**	**0.31×**
Hyundai Sonata	374	31.8%	2.12×
Kia Optima	232	37.5%	2.50×

> Our first method ranked on raw counts and penalised the Accord **for being popular**. Switching to share inverts the result — and independently surfaces the **Hyundai/Kia Theta II engine failure** episode from public data, without being told to look for it.

SAFETY

We attacked it on purpose

INJECTED INTO THE OUTPUT	CRITIC
Fake recall `NHTSA 19V888000`	CAUGHT
Plausible defect with no citation	CAUGHT
Dollar figure not in the cost table	CAUGHT
"All trims AWD V6, 45 mpg" (really FWD 4-cyl, 26–33)	CAUGHT
"worth *exactly* $11,000"	CAUGHT
Control: accurate output	NOT flagged

> **6/6, for $0.004.** The control is the one that matters: a Critic that flags everything would score 5/5 on attacks and be useless. It also caught an *omitted* real recall — a false negative, the more dangerous direction.

TOKENOMICS

One cent to analyse four cars

AGENT	TOKENS	COST
Dossier	3,044	$0.0011
Analyst	1,384	$0.0005
Critic	4,093	$0.0012
Per dossier	**8,522**	**$0.0028**

Mean of 3 real runs. gpt-6-luna via Portkey.

SCALE	LLM COST
1 session	$0.0091
100 users	$0.91
10,000 users	$91
10k MAU, 2×/mo	**$182/mo**

Session = 1 search + 3 dossiers.
Vehicle data adds **$0** — all keyless government APIs.

> **The Critic is 42% of run cost and the largest prompt.** That is the deliberate trade: half the budget on verification, for an app whose output moves thousands of dollars.

IMPACT

Does the number hold up?

Independent sanity check. AAA publishes **66.37¢/mile** for a *new* medium sedan. We compute **56.8¢/mile** for an eight-year-old used Accord — lower than new by a plausible margin, from a completely separate calculation path.

WHAT THE APP SURFACES	AMOUNT
Fuel spread hidden inside one model year	$2,750
Ownership cost vs purchase price	$34,100 vs $11,500
Maintenance alone vs purchase price	$12,877 vs $11,500

First outside user. Guessed **$30,000**; the app computes **$34,100** — she was within 12%. She still found it useful, and that is the point: the value was seeing *where the money goes*, not fixing her total. Then she asked whether I had known any of this **before we bought our own car here**. I had not.

Honestly: n=1, and the estimate came after she had seen the app, so it is anchored. Indicative, not a clean before-and-after.

LIMITS AND NEXT

What I would not claim

Known weaknesses

- Price model has **21% median error** — published, not buried. App shows a band, not a number.
- The `1.35×` out-of-warranty multiplier is the biggest judgement call, and it is an assumption.
- Share-based risk cannot see a defect an entire segment shares.
- The Critic is itself an LLM: a strong filter, not a proof.

Next, in order

1. Quantify user testing: before/after cost estimates.
2. Discount defects already remedied free under recall.
3. Cache dossiers per vehicle — collapses marginal cost.
4. Widen the depreciation fit beyond 581 listings.
5. VIN-level recall matching.

> **UsedCarDossier** — five free government data sources, zero API keys for vehicle data, every claim cited and machine-checked.
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